from a gun travels with a speed of 9 km per second.

Period of Revolution and Leap year

The period of time the earth takes to make one revolution around the sun determines the length of one year. The earth takes 365 days and 6 hours to complete one revolution. Earth takes 365.25 days to complete one trip around the Sun .That extra quarter of a day presents a challenge to our calendar system, which has one year as 365 days. To keep our yearly calendars consistent with our orbit around the Sun once in, every four years we add one day. The extra day added to is called a leap day, and the year the extra day is added to is called a leap year. The extra day is added to the month of February which has 29 days in a leap year.

Brain storming

How many birth days a person, whose life span supposed to be 60 years, would have seen in his/ her life time, if born on 29th February?

Effects of revolution of the earth

The revolution of the earth around the sun results in the following

- Cycle of seasons,
- Variation in length of days and nights,
- Variation in distribution of solar energy over the earth and the temperature zones.

2.15 Seasons

The seasons are caused due to the combined effect of the earth's revolution and the tilt of its axis to 23½° in the same direction throughout the year. In general, spring, summer, autumn and winter are the four seasons (Figure 2.12). The latitude at which the sun appears directly overhead changes as the earth orbits the sun. The sun appears to follow a yearly pattern of northward and southward motion in the sky, known as the 'apparent movement of the sun'. It gives an impression that the sun is continuously swinging north and south of the equator. Actually it is the earth that is moving around the sun on its tilted axis. It varies when observed on a daily and monthly basis, at different times of the year. On 21 March and 23 September the sun rises precisely in the east and sets exactly in the west.

Let us know!

How to calculate leap year? Take any year and divide by 4. If it is divisible (whole number with no reminder), it is a leap year.

Students' activity: calculate and identify the leap years from the following years

1992, 1995, 2000, 2005, 2008, 2010, 2012, 2014, 2017, 2020, 2024, 2030, 2035, 2040 and 2044.

Equinoxes and solstices

You already knew that the sunrays are vertical at noon. The vertical rays fall on a small area, giving more heat.

Equinoxes

Position of the earth on 21 March

Equinoxes occur when the earth reaches the points in its orbits where the equatorial and the orbital planes intersect, causing the sun to appear directly overhead at the equator. During the equinoxes the periods

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